

TRƯỜNG ĐẠI HỌC BÁCH KHOA - ĐẠI HỌC QUỐC GIA TP.HCM



BÀI THỰC HÀNH VI XỬ LÝ - VI ĐIỀU KHIỂN

LAB 1

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22/90/2026

Mục lục

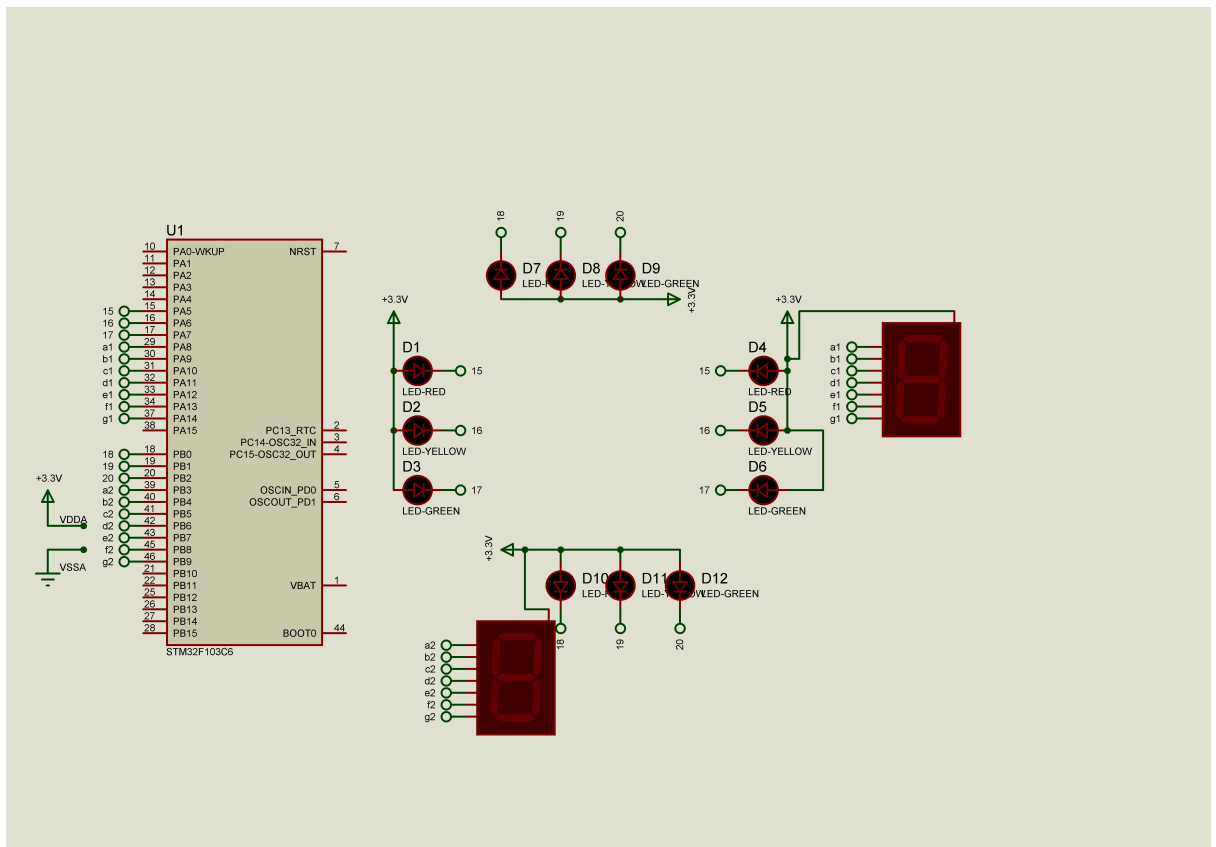
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Nội Dung Thực Hành

Code và Schematic có thể được xem trực tiếp và tải về tại GitHub Repository: Micro-controller Reports

1 Exercise 1 - 5:

- Schematic cho các bài từ 1 đến 5:



- Code bài 1:

Bật/tắt xen kẽ LED đỏ và LED vàng, mỗi trạng thái giữ 2 giây.

```

1 while (1)
2 {
3     HAL_GPIO_WritePin(LED_RED_GPIO_Port, LED_RED_Pin,
4                         GPIO_PIN_RESET);
5     HAL_GPIO_WritePin(LED_YELLOW_GPIO_Port, LED_YELLOW_Pin,
6                         GPIO_PIN_SET);
7     HAL_Delay(2000);
8     HAL_GPIO_WritePin(LED_RED_GPIO_Port, LED_RED_Pin,
9                         GPIO_PIN_SET);
10    HAL_GPIO_WritePin(LED_YELLOW_GPIO_Port, LED_YELLOW_Pin,
11                      GPIO_PIN_RESET);

```

```
8     HAL_Delay(2000);  
9 }
```

Listing 1: Exercise 1 — Toggle 2 LEDs

- **Code bài 2:**

Đèn giao thông 1 hướng: Đỏ 5s → Xanh 3s → Vàng 2s.

```
1 while (1) {  
2     // RED  
3     HAL_GPIO_WritePin(GPIOA, GPIO_PIN_5, GPIO_PIN_RESET);  
4     HAL_GPIO_WritePin(GPIOA, GPIO_PIN_6, GPIO_PIN_SET);  
5     HAL_GPIO_WritePin(GPIOA, GPIO_PIN_7, GPIO_PIN_SET);  
6     HAL_Delay(5000);  
7  
8     // GREEN  
9     HAL_GPIO_WritePin(GPIOA, GPIO_PIN_5, GPIO_PIN_SET);  
10    HAL_GPIO_WritePin(GPIOA, GPIO_PIN_6, GPIO_PIN_SET);  
11    HAL_GPIO_WritePin(GPIOA, GPIO_PIN_7, GPIO_PIN_RESET);  
12    HAL_Delay(3000);  
13  
14    // YELLOW  
15    HAL_GPIO_WritePin(GPIOA, GPIO_PIN_5, GPIO_PIN_SET);  
16    HAL_GPIO_WritePin(GPIOA, GPIO_PIN_6, GPIO_PIN_RESET);  
17    HAL_GPIO_WritePin(GPIOA, GPIO_PIN_7, GPIO_PIN_SET);  
18    HAL_Delay(2000);  
19 }
```

Listing 2: Exercise 2 — Traffic light 1 direction

- **Code bài 3:**

Đèn giao thông 2 hướng (COL/ROW) dùng hàm setGroupCOL và setGroupROW.

```
1 void setGroupCOL(int r, int y, int g) {  
2     HAL_GPIO_WritePin(GPIOA, LED_RED_COL_Pin,    r ?  
3         GPIO_PIN_RESET : GPIO_PIN_SET);  
4     HAL_GPIO_WritePin(GPIOA, LED_YELLOW_COL_Pin, y ?  
5         GPIO_PIN_RESET : GPIO_PIN_SET);  
6     HAL_GPIO_WritePin(GPIOA, LED_GREEN_COL_Pin,   g ?  
7         GPIO_PIN_RESET : GPIO_PIN_SET);  
8 }  
9  
10 void setGroupROW(int r, int y, int g) {  
11     HAL_GPIO_WritePin(GPIOB, LED_RED_ROW_Pin,    r ?  
12         GPIO_PIN_RESET : GPIO_PIN_SET);  
13     HAL_GPIO_WritePin(GPIOB, LED_YELLOW_ROW_Pin, y ?  
14         GPIO_PIN_RESET : GPIO_PIN_SET);  
15     HAL_GPIO_WritePin(GPIOB, LED_GREEN_ROW_Pin,   g ?  
16         GPIO_PIN_RESET : GPIO_PIN_SET);  
17 }
```

Listing 3: Exercise 3 — Helper functions

```

1 while (1)
2 {
3     setGroupCOL(1,0,0); setGroupROW(0,0,1); HAL_Delay(3000);
4     setGroupCOL(1,0,0); setGroupROW(0,1,0); HAL_Delay(2000);
5     setGroupCOL(0,0,1); setGroupROW(1,0,0); HAL_Delay(3000);
6     setGroupCOL(0,1,0); setGroupROW(1,0,0); HAL_Delay(2000);
7 }

```

Listing 4: Exercise 3 — Main loop (2-direction traffic light)

- **Code bài 4:**

Hiển thị đếm 0–9 trên LED 7 đoạn.

```

1 uint8_t segTable[10][7] = {
2     // a  b  c  d  e  f  g
3     { 0, 0, 0, 0, 0, 0, 1 }, // 0
4     { 1, 0, 0, 1, 1, 1, 1 }, // 1
5     { 0, 0, 1, 0, 0, 1, 0 }, // 2
6     { 0, 0, 0, 0, 1, 1, 0 }, // 3
7     { 1, 0, 0, 1, 1, 0, 0 }, // 4
8     { 0, 1, 0, 0, 1, 0, 0 }, // 5
9     { 0, 1, 0, 0, 0, 0, 0 }, // 6
10    { 0, 0, 0, 1, 1, 1, 1 }, // 7
11    { 0, 0, 0, 0, 0, 0, 0 }, // 8
12    { 0, 0, 0, 0, 1, 0, 0 }  // 9
13 };
14
15 void display7SEG(int num) {
16     HAL_GPIO_WritePin(GPIOA, a1_Pin, segTable[num][0]);
17     HAL_GPIO_WritePin(GPIOA, b1_Pin, segTable[num][1]);
18     HAL_GPIO_WritePin(GPIOA, c1_Pin, segTable[num][2]);
19     HAL_GPIO_WritePin(GPIOA, d1_Pin, segTable[num][3]);
20     HAL_GPIO_WritePin(GPIOA, e1_Pin, segTable[num][4]);
21     HAL_GPIO_WritePin(GPIOA, f1_Pin, segTable[num][5]);
22     HAL_GPIO_WritePin(GPIOA, g1_Pin, segTable[num][6]);
23 }

```

Listing 5: Exercise 4 — 7-segment display function and lookup table

```

1 int counter = 0;
2 while (1)
3 {
4     if (counter >= 10) counter = 0;
5     display7SEG(counter++);
6     HAL_Delay(1000);
7 }

```

Listing 6: Exercise 4 — Main loop (count 0–9)

- **Code bài 5:**

Đèn giao thông 2 hướng kết hợp 2 LED 7 đoạn đếm ngược.

```
1 void setGroupCOL(int r, int y, int g) {
2     HAL_GPIO_WritePin(GPIOA, LED_RED_COL_Pin,      r ?
3         GPIO_PIN_RESET : GPIO_PIN_SET);
4     HAL_GPIO_WritePin(GPIOA, LED_YELLOW_COL_Pin, y ?
5         GPIO_PIN_RESET : GPIO_PIN_SET);
6     HAL_GPIO_WritePin(GPIOA, LED_GREEN_COL_Pin,  g ?
7         GPIO_PIN_RESET : GPIO_PIN_SET);
8 }
9
10 void setGroupROW(int r, int y, int g) {
11     HAL_GPIO_WritePin(GPIOB, LED_RED_ROW_Pin,      r ?
12         GPIO_PIN_RESET : GPIO_PIN_SET);
13     HAL_GPIO_WritePin(GPIOB, LED_YELLOW_ROW_Pin, y ?
14         GPIO_PIN_RESET : GPIO_PIN_SET);
15     HAL_GPIO_WritePin(GPIOB, LED_GREEN_ROW_Pin,  g ?
16         GPIO_PIN_RESET : GPIO_PIN_SET);
17 }
18
19 uint8_t segTable[10][7] = {
20     { 0, 0, 0, 0, 0, 0, 1 }, // 0
21     { 1, 0, 0, 1, 1, 1, 1 }, // 1
22     { 0, 0, 1, 0, 0, 1, 0 }, // 2
23     { 0, 0, 0, 0, 1, 1, 0 }, // 3
24     { 1, 0, 0, 1, 1, 0, 0 }, // 4
25     { 0, 1, 0, 0, 1, 0, 0 }, // 5
26     { 0, 1, 0, 0, 0, 0, 0 }, // 6
27     { 0, 0, 0, 1, 1, 1, 1 }, // 7
28     { 0, 0, 0, 0, 0, 0, 0 }, // 8
29     { 0, 0, 0, 0, 1, 0, 0 }, // 9
30 };
31
32 void display7SEG_COL(int num) {
33     HAL_GPIO_WritePin(GPIOA, a1_Pin, segTable[num][0]);
34     HAL_GPIO_WritePin(GPIOA, b1_Pin, segTable[num][1]);
35     HAL_GPIO_WritePin(GPIOA, c1_Pin, segTable[num][2]);
36     HAL_GPIO_WritePin(GPIOA, d1_Pin, segTable[num][3]);
37     HAL_GPIO_WritePin(GPIOA, e1_Pin, segTable[num][4]);
38     HAL_GPIO_WritePin(GPIOA, f1_Pin, segTable[num][5]);
39     HAL_GPIO_WritePin(GPIOA, g1_Pin, segTable[num][6]);
40 }
41
42 void display7SEG_ROW(int num) {
43     HAL_GPIO_WritePin(GPIOB, a2_Pin, segTable[num][0]);
44     HAL_GPIO_WritePin(GPIOB, b2_Pin, segTable[num][1]);
45     HAL_GPIO_WritePin(GPIOB, c2_Pin, segTable[num][2]);
46     HAL_GPIO_WritePin(GPIOB, d2_Pin, segTable[num][3]);
47     HAL_GPIO_WritePin(GPIOB, e2_Pin, segTable[num][4]);
48     HAL_GPIO_WritePin(GPIOB, f2_Pin, segTable[num][5]);
49     HAL_GPIO_WritePin(GPIOB, g2_Pin, segTable[num][6]);
50 }
```

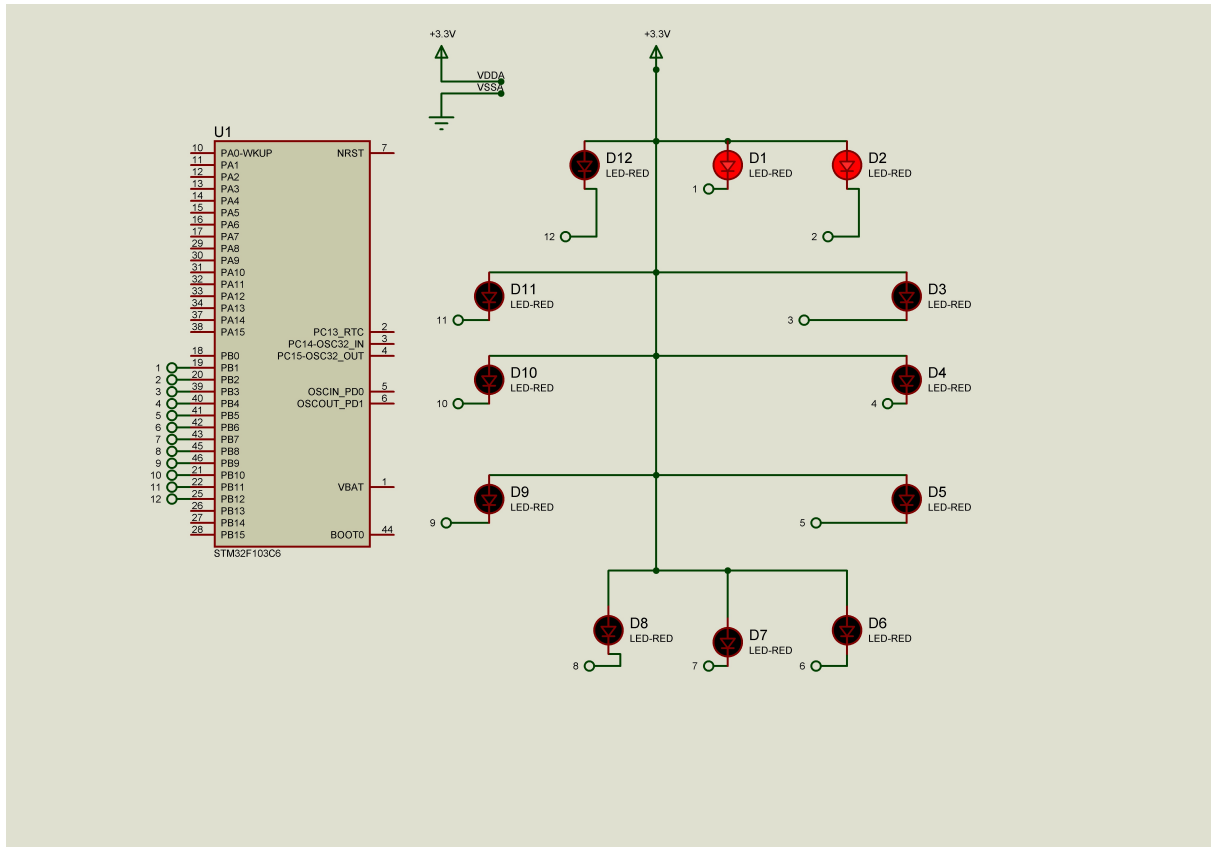
Listing 7: Exercise 5 — Helper functions and 7-seg display

```
1  int t = 0;
2  while (1)
3  {
4      if (t < 5) {
5          setGroupCOL(1, 0, 0);
6          setGroupROW(0, t >= 3, t < 3);
7          display7SEG_COL(5 - t);
8          display7SEG_ROW(t < 3 ? 3 - t : 5 - t);
9      } else {
10         setGroupCOL(0, t >= 8, t < 8);
11         setGroupROW(1, 0, 0);
12         display7SEG_COL(t < 8 ? 8 - t : 10 - t);
13         display7SEG_ROW(10 - t);
14     }
15     HAL_Delay(1000);
16     t++;
17     if (t >= 10) t = 0;
18 }
```

Listing 8: Exercise 5 — Main loop (2-direction with countdown)

2 Exercise 6 - 10:

- Schematic cho các bài từ 6 đến 10:



Mảng chân dùng chung cho bài 6–10:

Tất cả các bài 6–10 sử dụng chung mảng ánh xạ 12 vị trí đồng hồ sang chân GPIO trên GPIOB:

```

1 uint16_t clockPins[12] = {
2     a1_Pin, a2_Pin, a3_Pin,
3     b1_Pin, b2_Pin, b3_Pin,
4     c1_Pin, c2_Pin, c3_Pin,
5     d1_Pin, d2_Pin, d3_Pin
6 };

```

Listing 9: Clock pins mapping (shared by Ex.6–10)

• Code bài 6:

Quét lần lượt 12 LED theo vòng tròn đồng hồ, mỗi LED sáng 500ms rồi tắt.

```

1 while (1)
2 {
3     for (int i = 0; i < 12; i++) {
4         HAL_GPIO_WritePin(GPIOB, clockPins[i], GPIO_PIN_RESET);
5         HAL_Delay(500);
6         HAL_GPIO_WritePin(GPIOB, clockPins[i], GPIO_PIN_SET);
7     }
8 }

```

Listing 10: Exercise 6 — Scanning clock LEDs

- **Code bài 7:**

Hàm `clearAllClock` tắt tất cả 12 LED cùng lúc. Demo: bật 9 LED, đợi 2s, rồi tắt hết.

```
1 void clearAllClock(void) {  
2     HAL_GPIO_WritePin(GPIOB,  
3         a1_Pin | a2_Pin | a3_Pin |  
4         b1_Pin | b2_Pin | b3_Pin |  
5         c1_Pin | c2_Pin | c3_Pin |  
6         d1_Pin | d2_Pin | d3_Pin,  
7         GPIO_PIN_SET);  
8 }
```

Listing 11: Exercise 7 — `clearAllClock` function

```
1 while (1)  
2 {  
3     HAL_GPIO_WritePin(GPIOB,  
4         a1_Pin | a2_Pin | a3_Pin |  
5         b1_Pin | b2_Pin | b3_Pin |  
6         c1_Pin | c2_Pin | c3_Pin, GPIO_PIN_SET);  
7     HAL_Delay(2000);  
8  
9     clearAllClock();  
10    HAL_Delay(2000);  
11 }
```

Listing 12: Exercise 7 — Main loop demo

- **Code bài 8:**

Hàm `setNumberOnClock` bật 1 LED tại vị trí chỉ định. Demo: xóa hết rồi bật vị trí 0, đợi 1s, xóa rồi bật vị trí 3.

```
1 void setNumberOnClock(int num) {  
2     HAL_GPIO_WritePin(GPIOB, clockPins[num], GPIO_PIN_RESET);  
3 }
```

Listing 13: Exercise 8 — `setNumberOnClock` function

```
1 while (1)  
2 {  
3     clearAllClock();  
4     setNumberOnClock(0);  
5     HAL_Delay(1000);  
6     clearAllClock();  
7     setNumberOnClock(3);  
8     HAL_Delay(1000);  
9 }
```

Listing 14: Exercise 8 — Main loop demo

• Code bài 9:

Hàm `clearNumberOnClock` tắt 1 LED tại vị trí chỉ định. Thêm hàm `activeAllClock` bật tất cả. Demo: bật hết, đợi 2s, tắt vị trí 3, đợi 5s.

```
1 void activeAllClock(void) {
2     HAL_GPIO_WritePin(GPIOB,
3         a1_Pin | a2_Pin | a3_Pin |
4         b1_Pin | b2_Pin | b3_Pin |
5         c1_Pin | c2_Pin | c3_Pin |
6         d1_Pin | d2_Pin | d3_Pin,
7         GPIO_PIN_RESET);
8 }
9
10 void clearNumberOnClock(int num) {
11     HAL_GPIO_WritePin(GPIOB, clockPins[num], GPIO_PIN_SET);
12 }
```

Listing 15: Exercise 9 — `clearNumberOnClock` and `activeAllClock` functions

```
1 while (1)
2 {
3     activeAllClock();
4     HAL_Delay(2000);
5     clearNumberOnClock(3);
6     HAL_Delay(5000);
7 }
```

Listing 16: Exercise 9 — Main loop demo

• Code bài 10:

Đồng hồ analog: dùng hàm `updateClockDisplay` hiển thị đồng thời kim giờ, kim phút (chia 5), kim giây (chia 5) trên 12 LED.

```
1 int hour = 0, minute = 0, second = 0;
2
3 void updateClockDisplay(void) {
4     int h = hour;
5     int m = minute / 5;
6     int s = second / 5;
7
8     clearAllClock();
9     setNumberOnClock(h);
10    setNumberOnClock(m);
11    setNumberOnClock(s);
12 }
```

Listing 17: Exercise 10 — `updateClockDisplay` function

```
1 while (1)
2 {
3     updateClockDisplay();
4     HAL_Delay(100);
5 }
```

```
5
6     second++;
7     if (second >= 60) {
8         second = 0;
9         minute++;
10        if (minute >= 60) {
11            minute = 0;
12            hour++;
13            if (hour >= 12) hour = 0;
14        }
15    }
16 }
```

Listing 18: Exercise 10 — Main loop (analog clock)